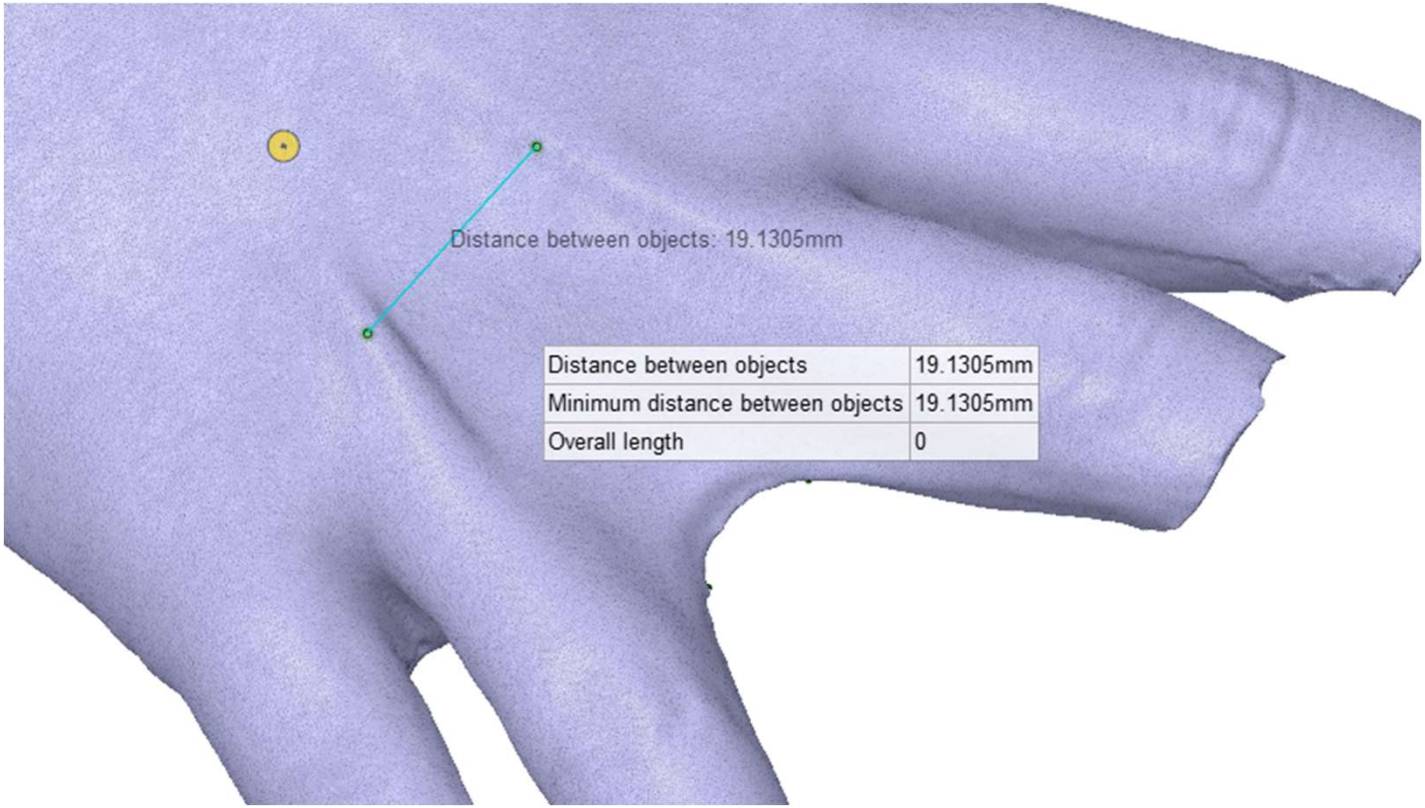
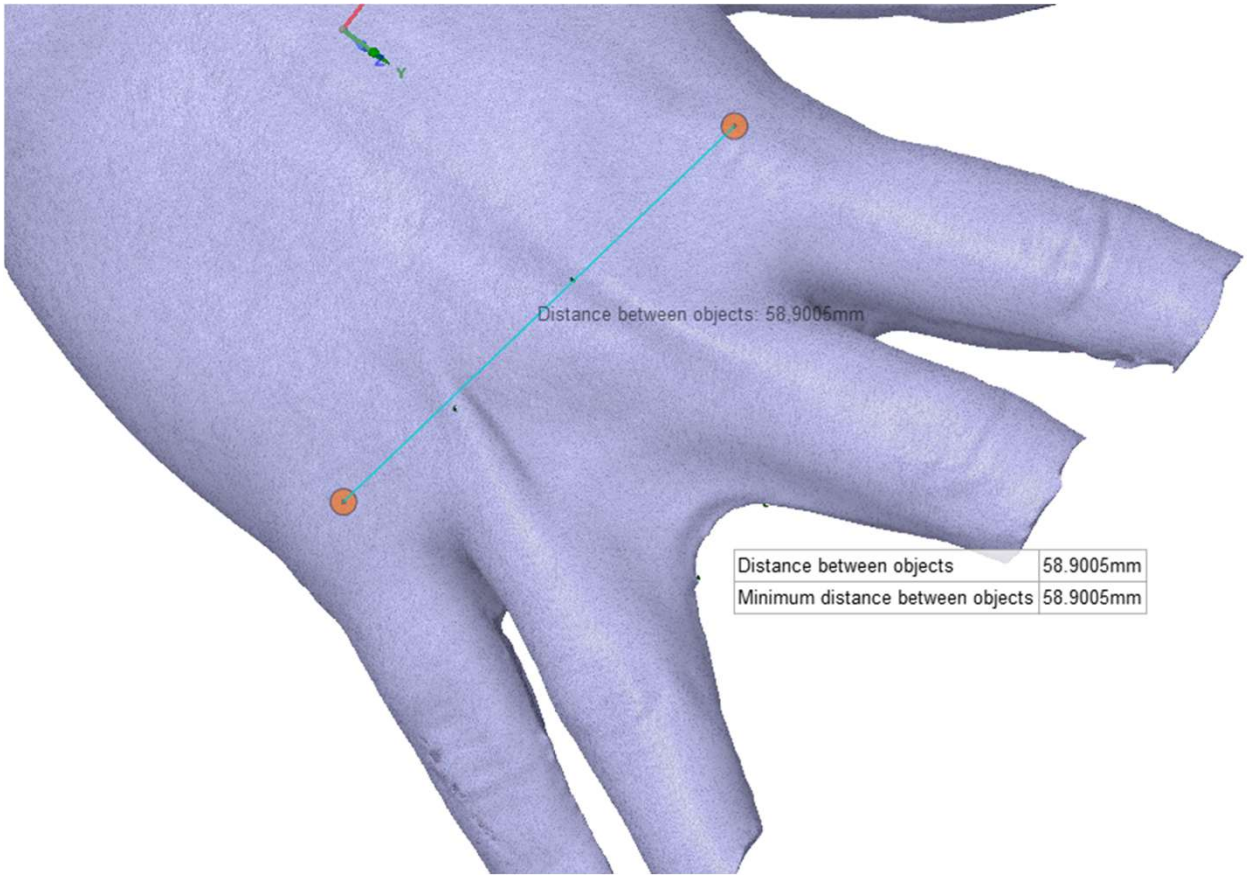
Research

Yewon Jang

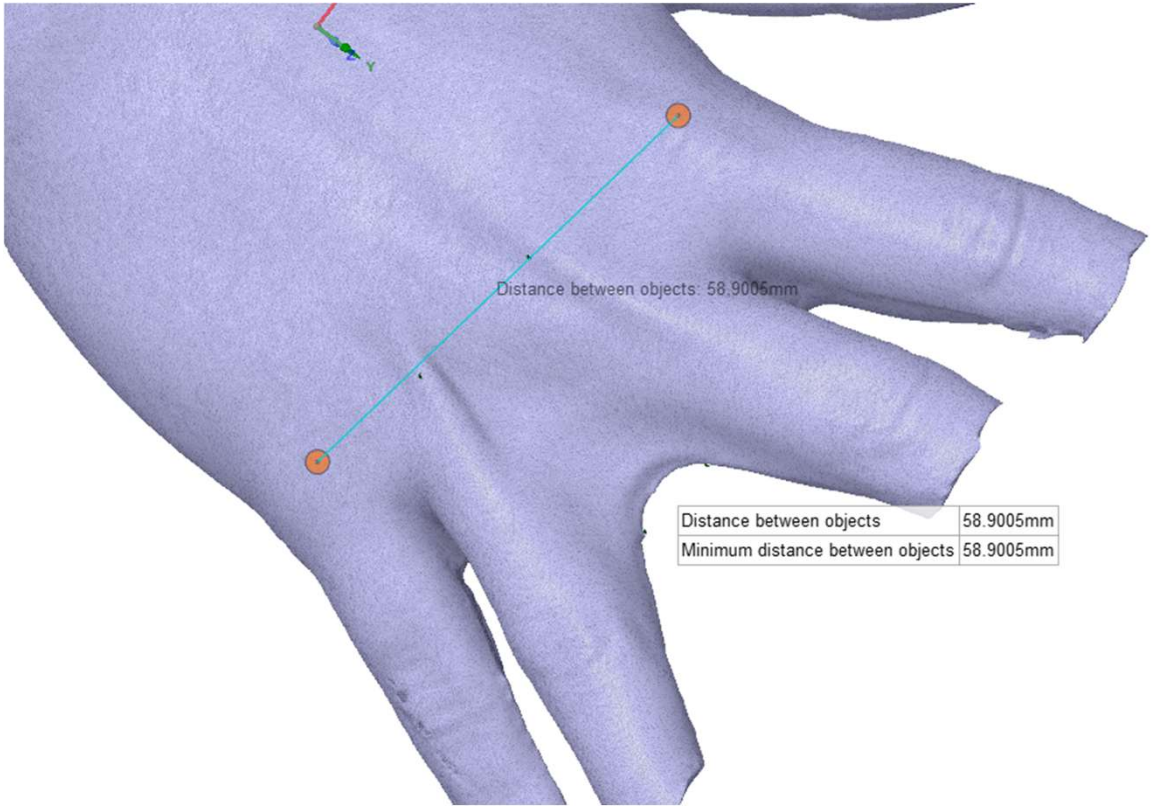
Measurements



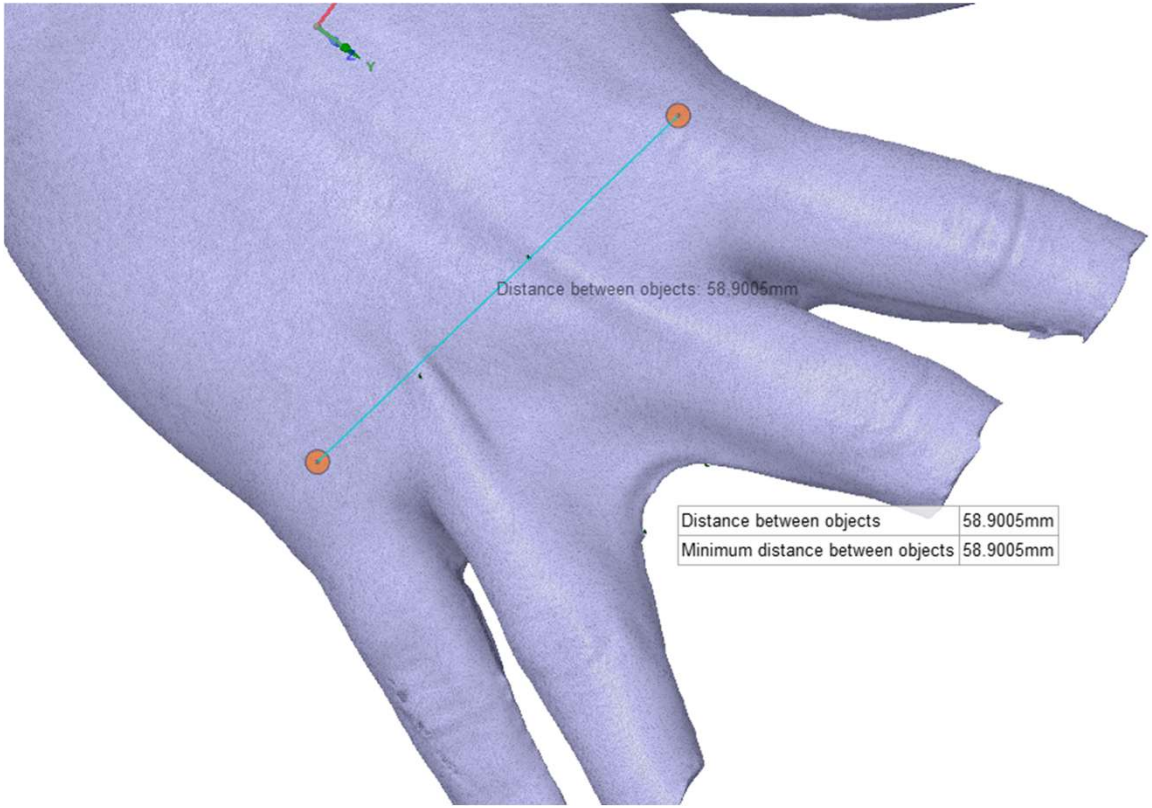
Measurements



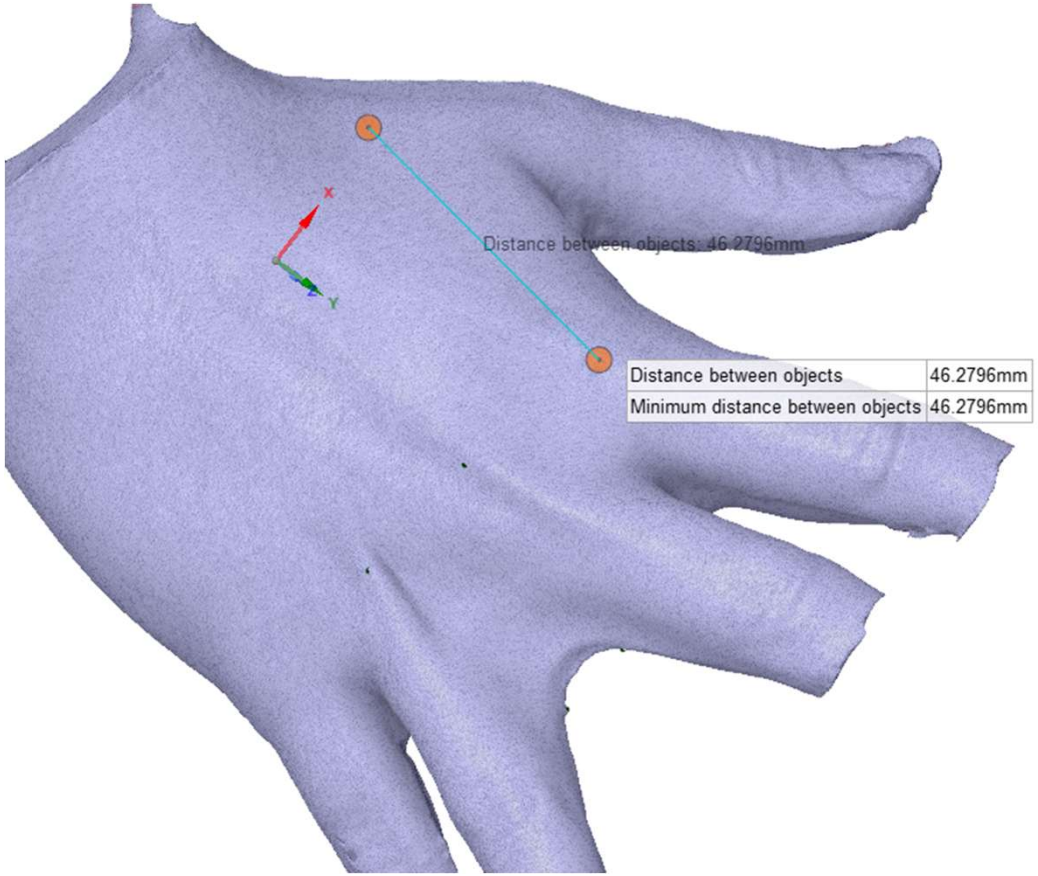
Measurements



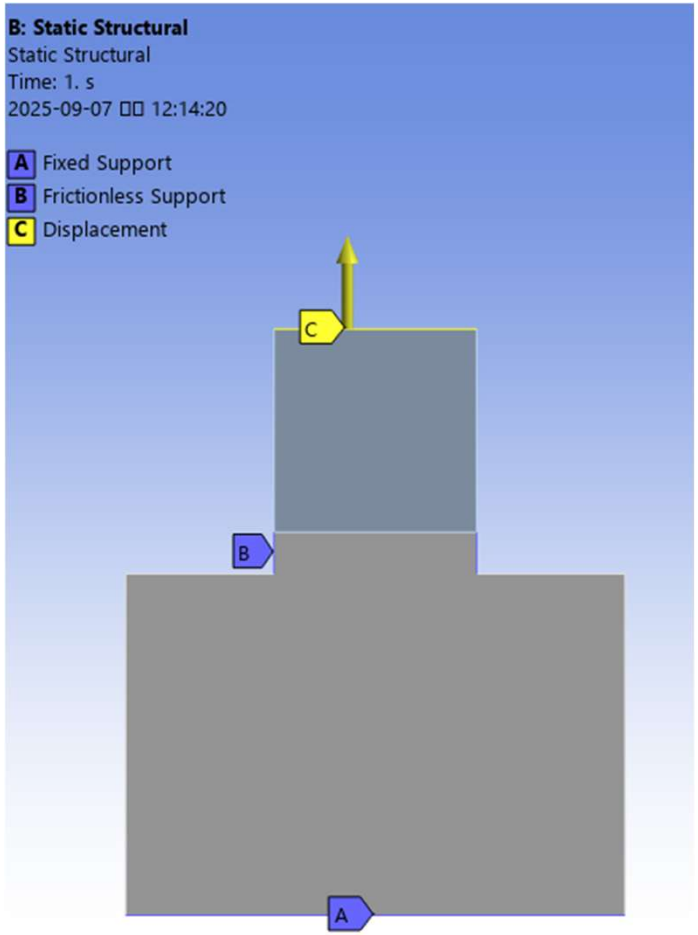
Measurements



Measurements



Settings



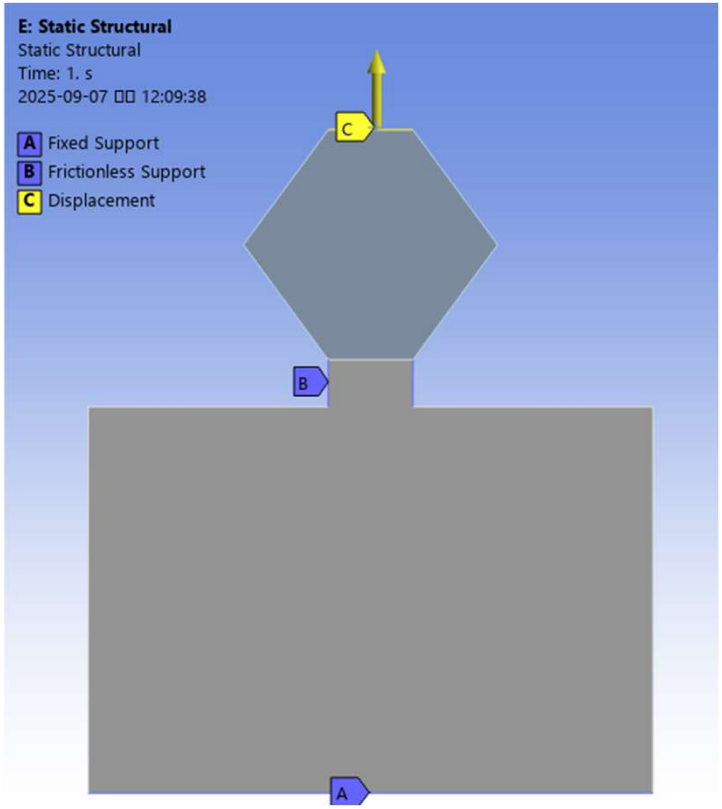
Details of "Displacement"

Scope	
Scoping Method	Geometry Selection
Geometry	1 Edge
Definition	
Type	Displacement
Define By	Components
Coordinate System	Global Coordinate System
X Component	Free
Y Component	16.4 mm (ramped)
Suppressed	No

Properties of Outline Row 3: Skin

	A	B	C	D	E
1	Property	Value	Unit		
2	Material Field Variables	Table			
3	Density	1100	kg m^-3		
4	Isotropic Secant Coefficient of Thermal Expansion				
5	Coefficient of Thermal Expansion	0.0001	C^-1		
6	Isotropic Elasticity				
7	Derive from	Young's...			
8	Young's Modulus	8E+05	Pa		
9	Poisson's Ratio	0.47			
10	Bulk Modulus	4.4444E+06	Pa		
11	Shear Modulus	2.7211E+05	Pa		

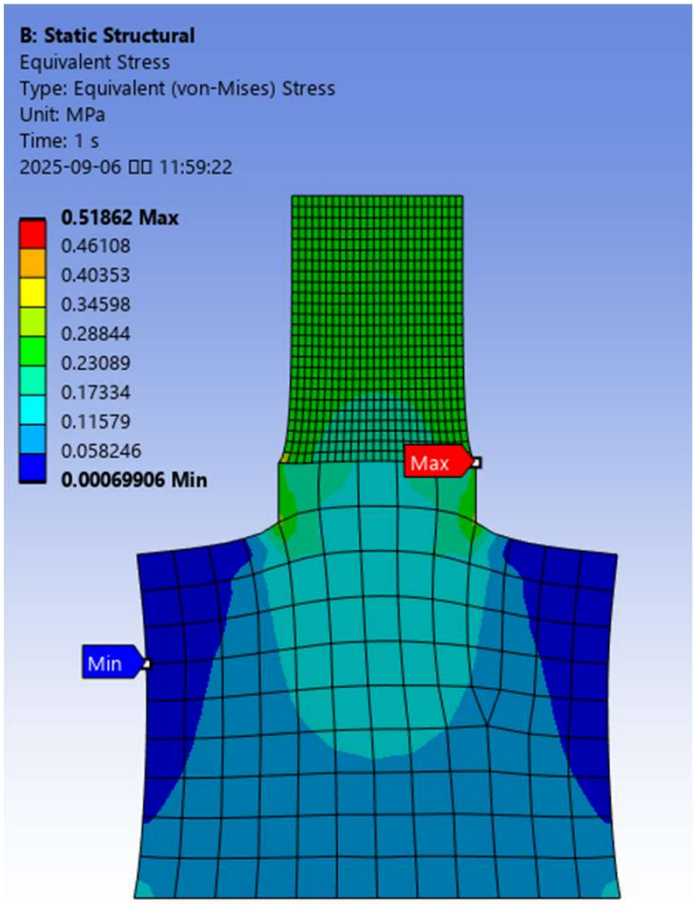
Settings

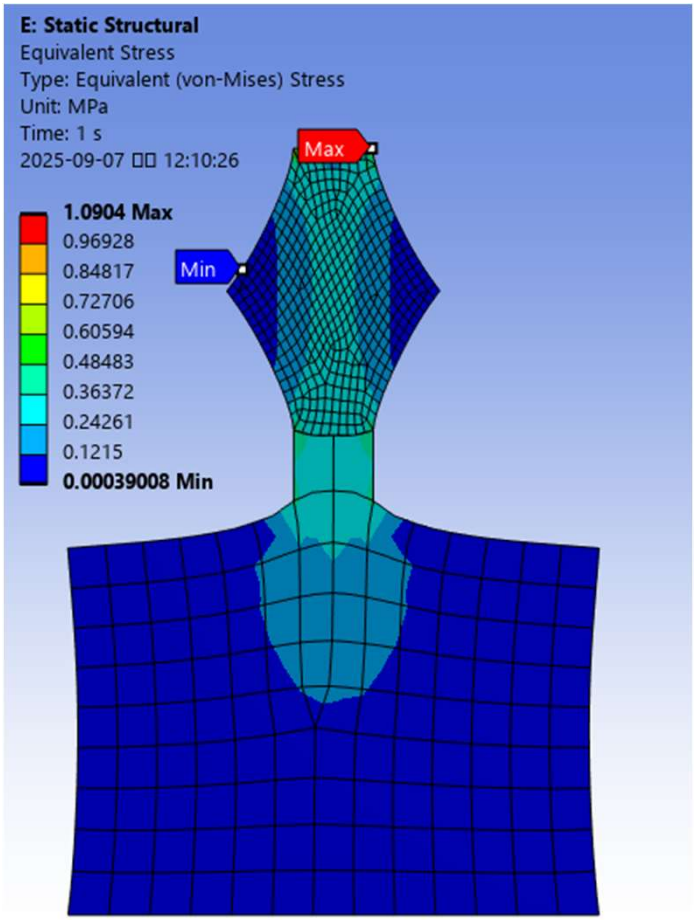


Details of "Displacement"

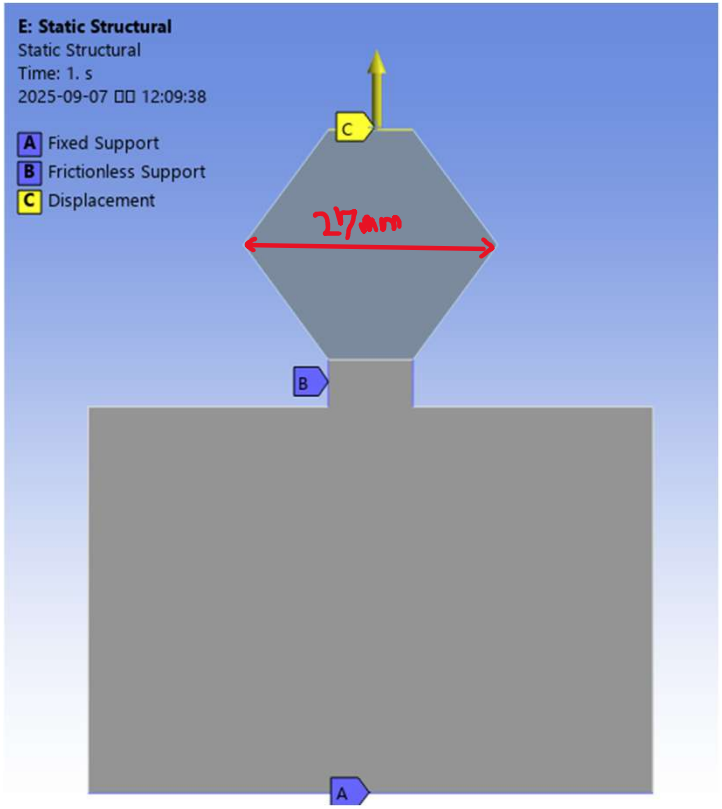
Scope	
Scoping Method	Geometry Selection
Geometry	1 Edge
Definition	
Type	Displacement
Define By	Components
Coordinate System	Global Coordinate System
X Component	Free
☐ Y Component	16.4 mm (ramped)
Suppressed	No

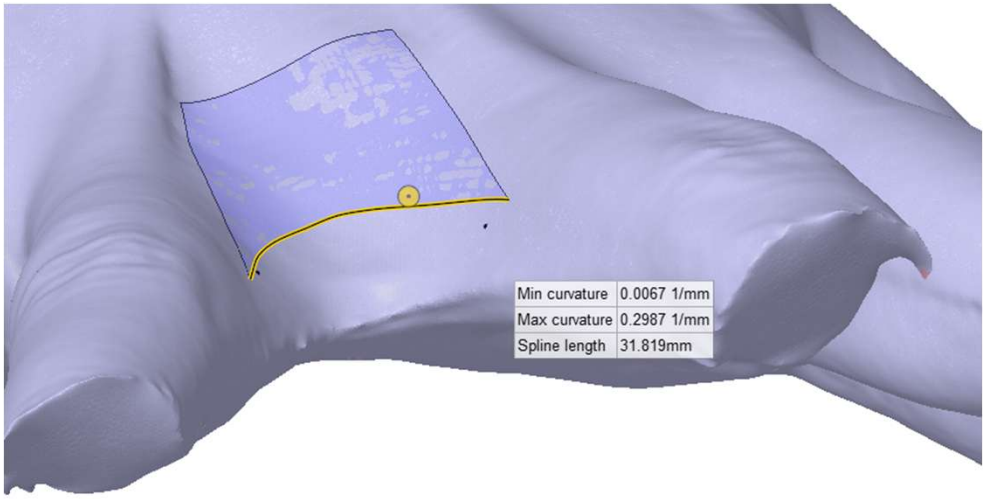
Results


B: Static Structural
Equivalent Stress
Type: Equivalent (von-Mises) Stress
Unit: MPa
Time: 1 s
2025-09-06 □□ 11:59:22
0.51862 Max
0.46108
0.40353
0.34598
0.28844
0.23089
0.17334
0.11579
0.058246
0.00069906 Min
Max
Min


E: Static Structural
Equivalent Stress
Type: Equivalent (von-Mises) Stress
Unit: MPa
Time: 1 s
2025-09-07 □□ 12:10:26
1.0904 Max
0.96928
0.84817
0.72706
0.60594
0.48483
0.36372
0.24261
0.1215
0.00039008 Min
Max
Min

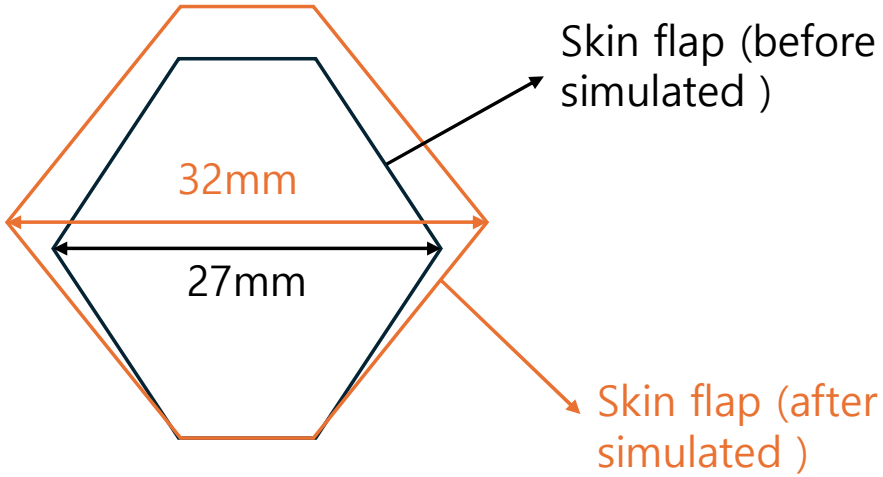
Realistic model



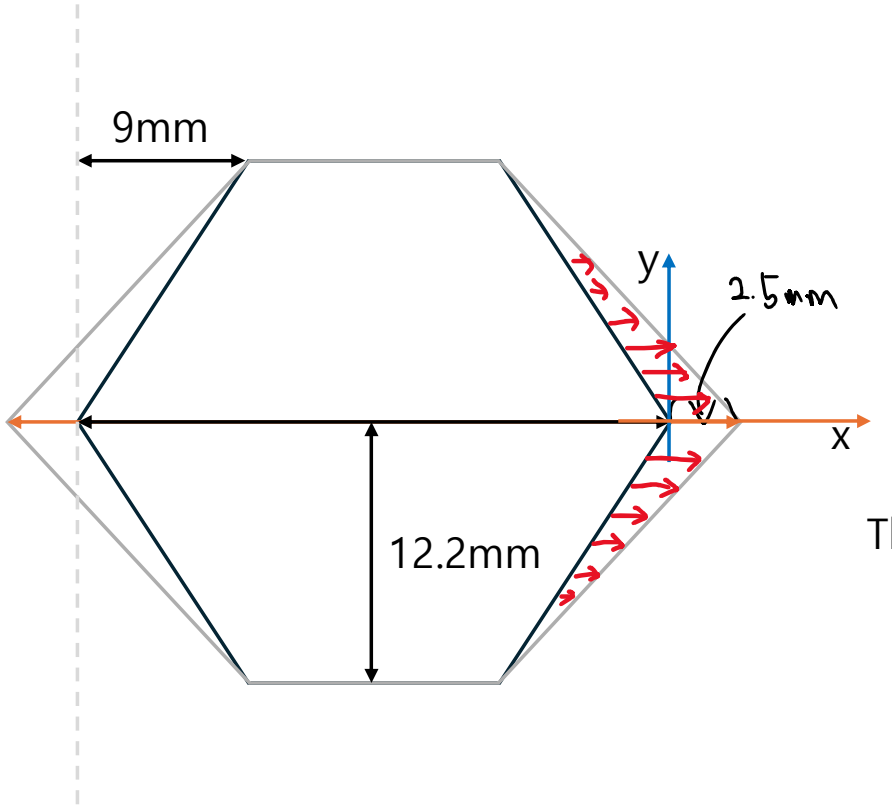


The spine length is around 32mm.

Displacement



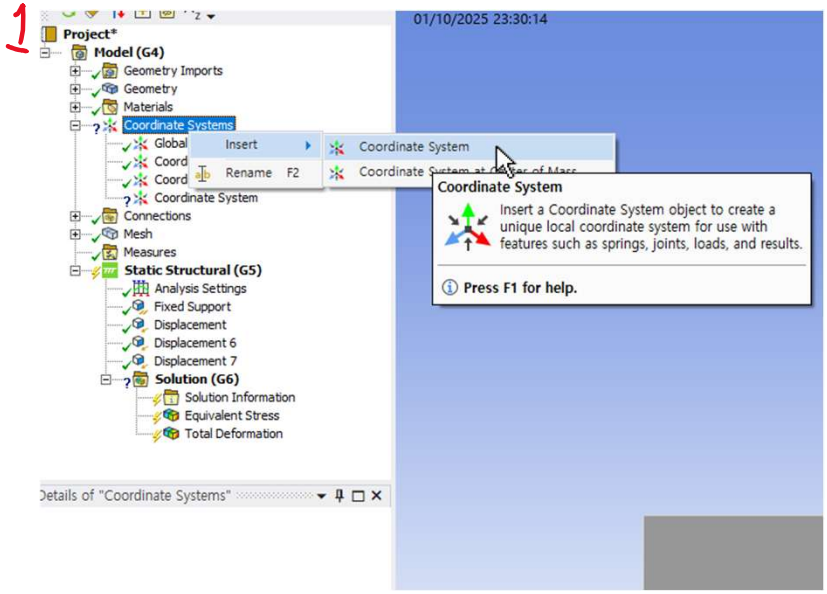
Displacement

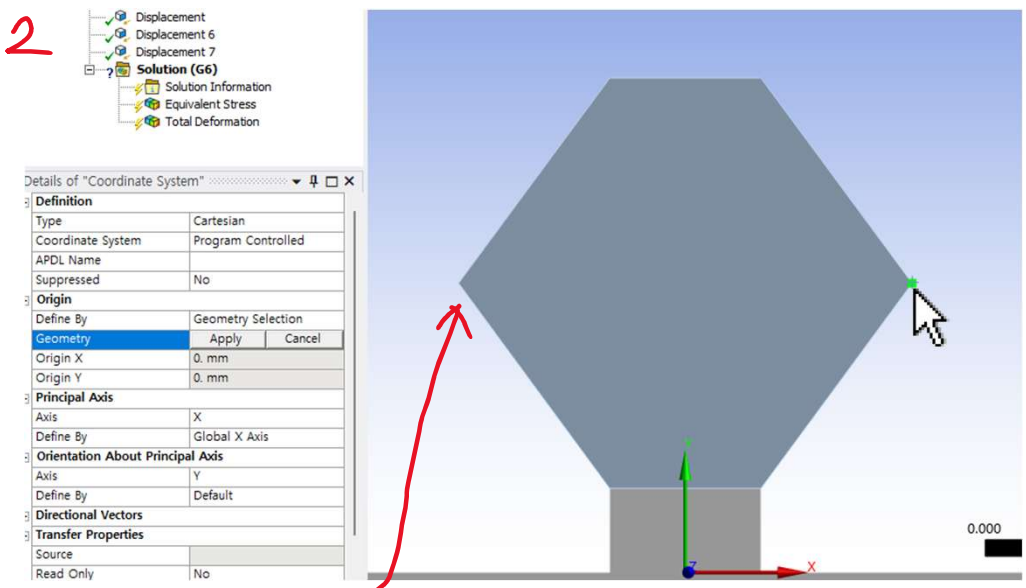


The displacement along X axis = 2.5/9 (x+9)

How to set the displacement

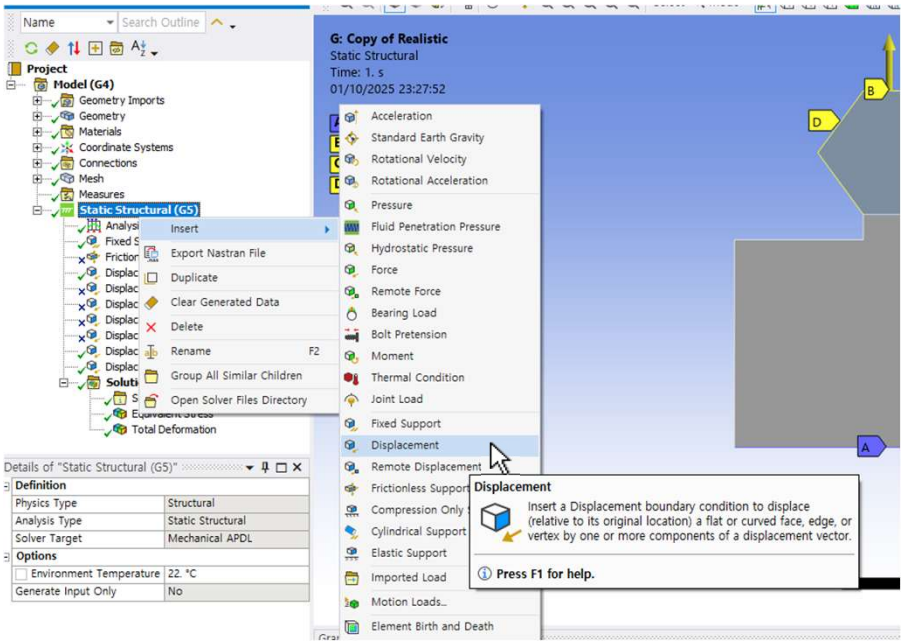
1. Insert a Coordinate system on the both side of hexagon

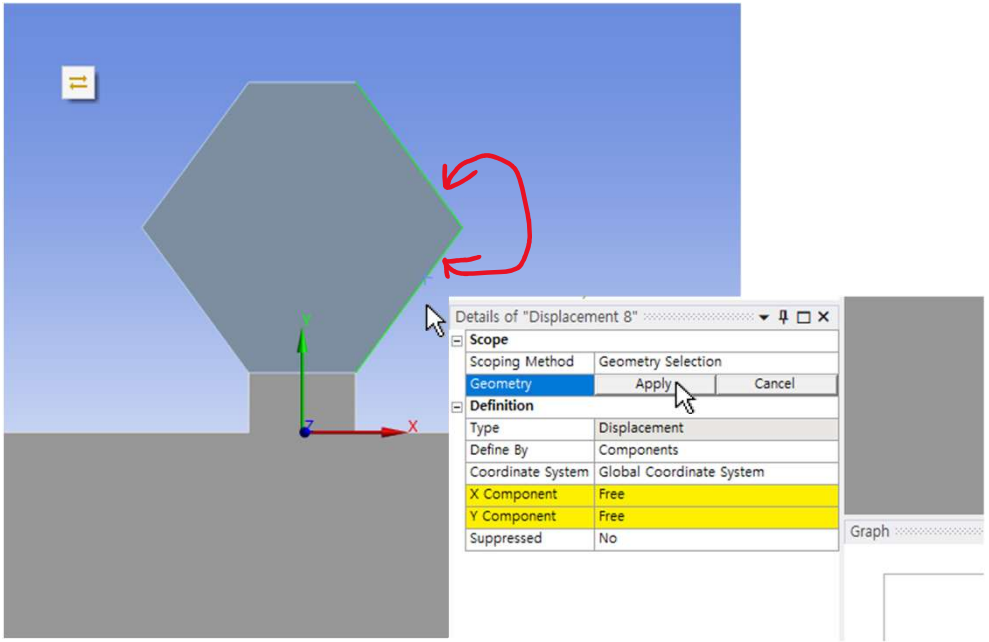




2. Do the same thing on the other side

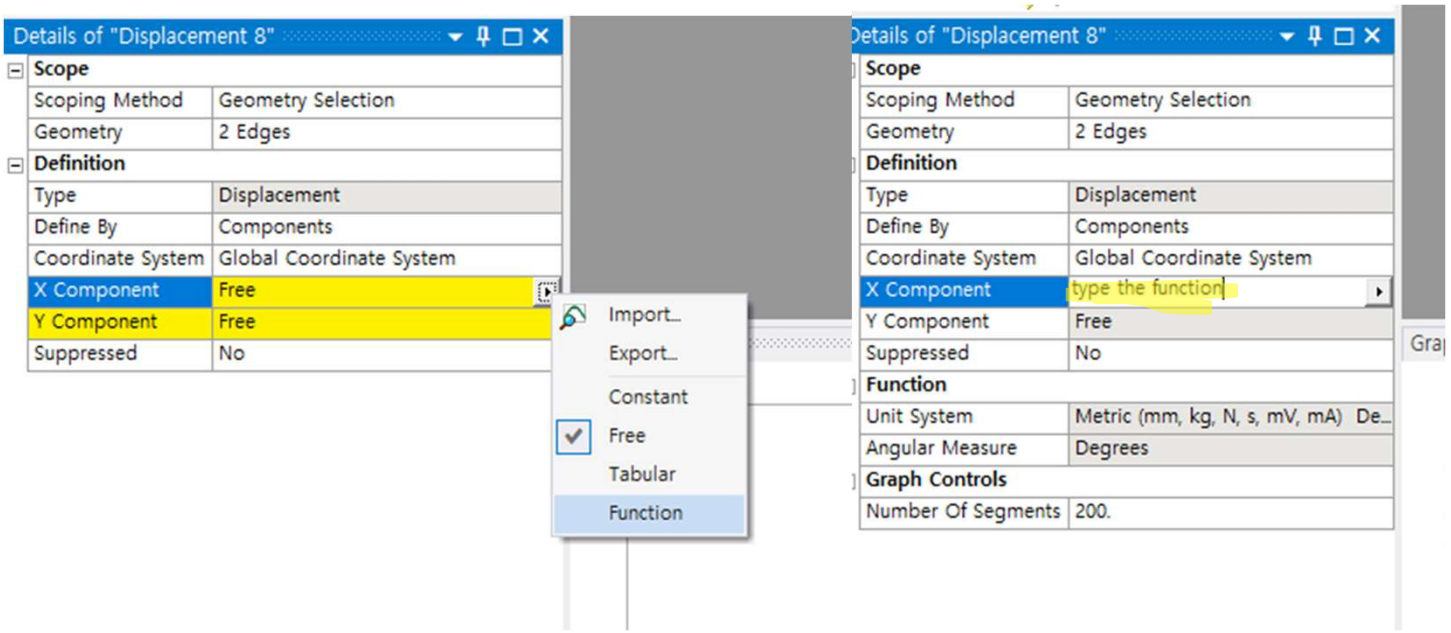
How to set the displacement





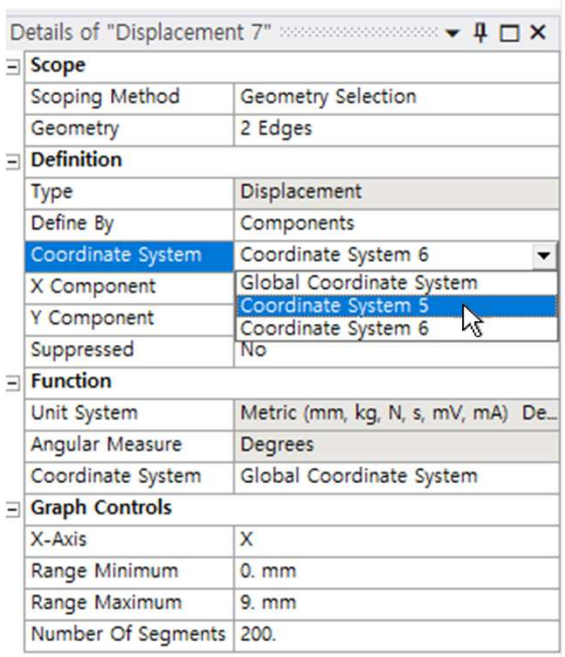
Apply two edges on the hexagon

How to set the displacement



Choose 'function' and write the function that we derive previous in the box

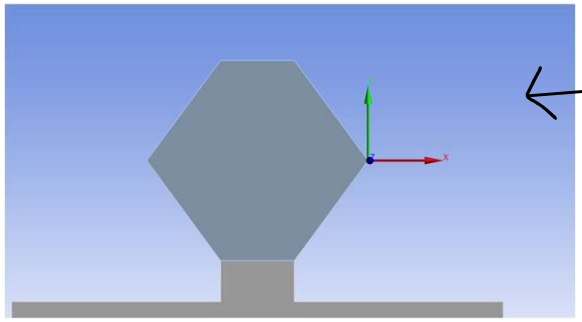
How to set the displacement



Choose the proper coordinate system

Do the same progress on the other side of the hexagon

Displacement

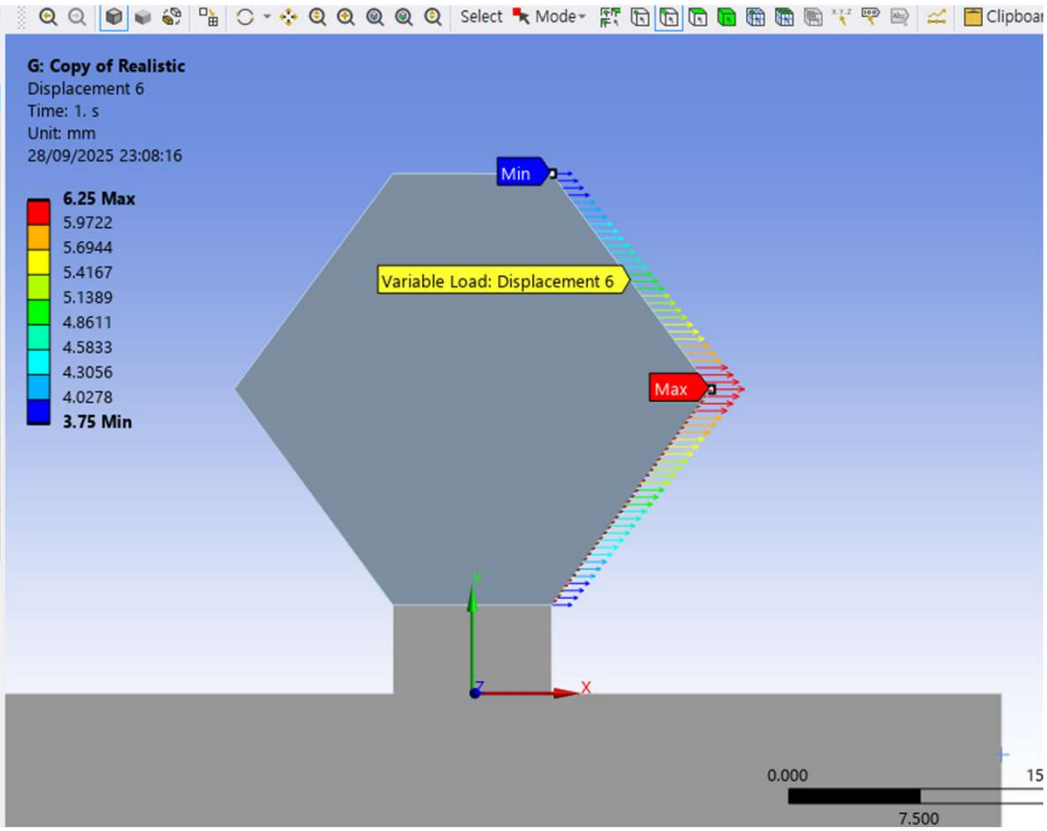


Outline

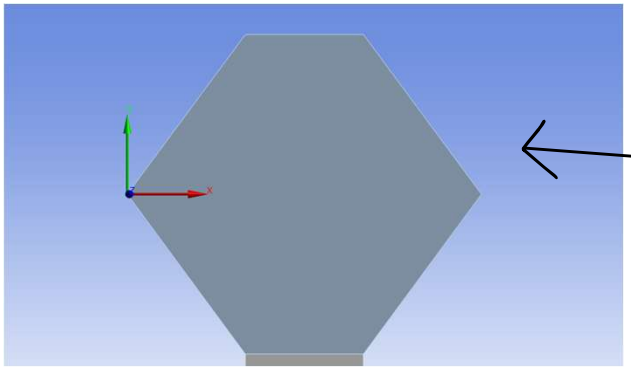
Name | Search Outline

Details of "Displacement 6"

Scope	
Scoping Method	Geometry Selection
Geometry	2 Edges
Definition	
Type	Displacement
Define By	Components
Coordinate System	Coordinate System 5
X Component	= (2.5 / 9) * (x + 9)
Y Component	Free
Suppressed	No
Function	
Unit System	Metric (mm, kg, N, s, mV, mA) De...
Angular Measure	Degrees
Coordinate System	Global Coordinate System
Graph Controls	
X-Axis	X
Range Minimum	-9. mm
Range Maximum	0. mm
Number Of Segments	200.



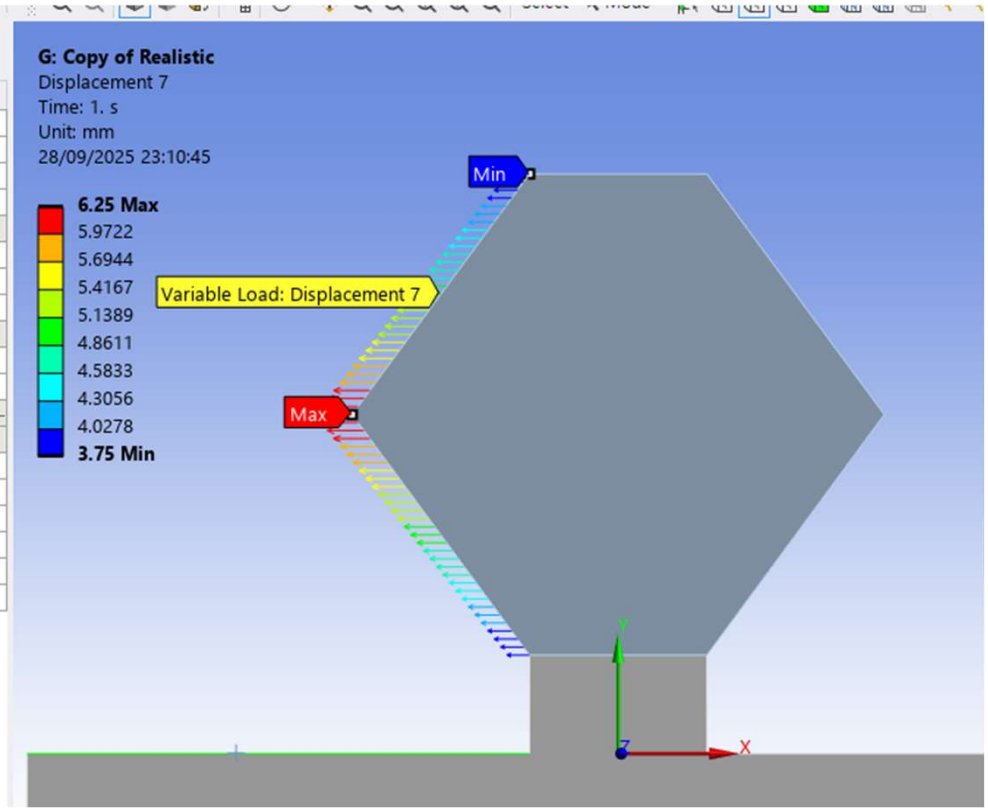
Displacement (Another side)



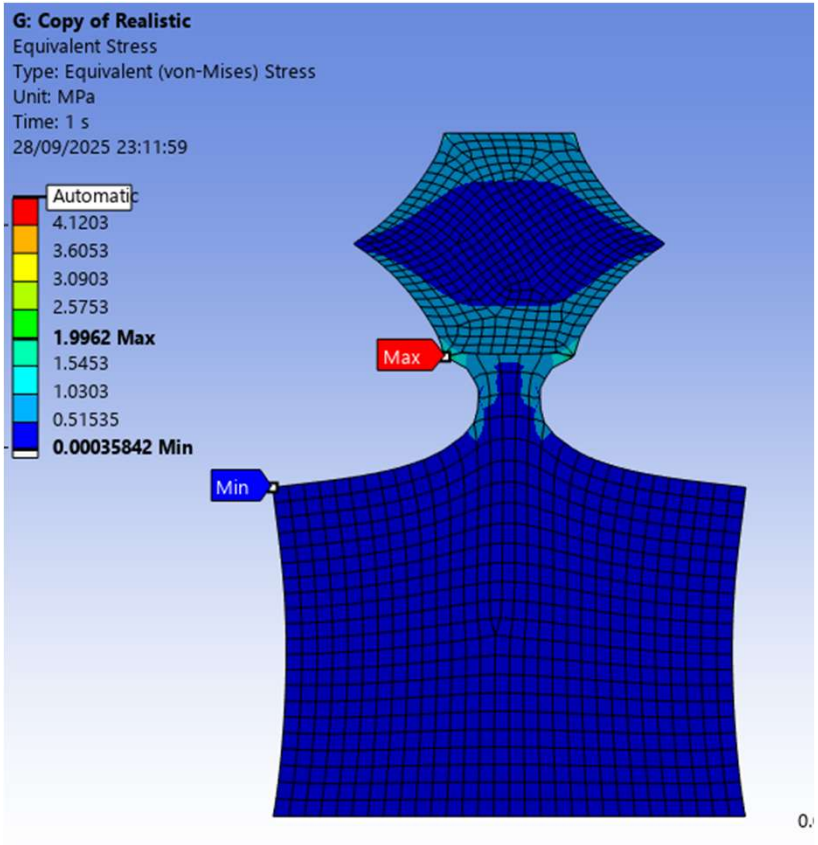
Name | Search Outline

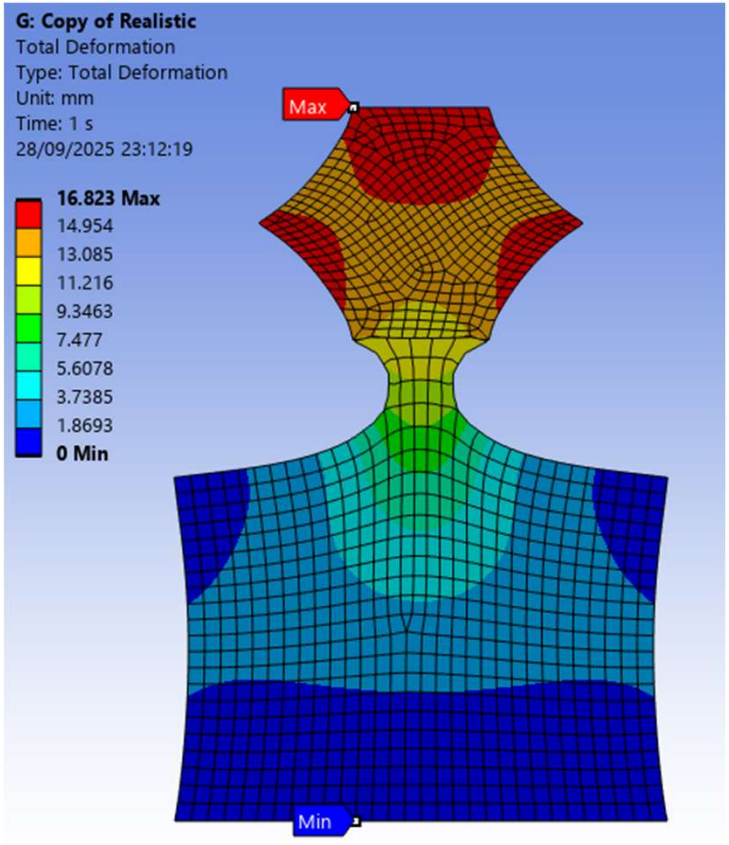
Details of "Displacement 7"

Scope	
Scoping Method	Geometry Selection
Geometry	2 Edges
Definition	
Type	Displacement
Define By	Components
Coordinate System	Coordinate System 6
X Component	= (2.5 / 9) * (x-9)
Y Component	Free
Suppressed	No
Function	
Unit System	Metric (mm, kg, N, s, mV, mA) De...
Angular Measure	Degrees
Coordinate System	Global Coordinate System
Graph Controls	
X-Axis	X
Range Minimum	0. mm
Range Maximum	9. mm
Number Of Segments	200.



Result


G: Copy of Realistic
Equivalent Stress
Type: Equivalent (von-Mises) Stress
Unit: MPa
Time: 1 s
28/09/2025 23:11:59
Automatic
4.1203
3.6053
3.0903
2.5753
1.9962 Max
1.5453
1.0303
0.51535
0.00035842 Min
Max
Min


G: Copy of Realistic
Total Deformation
Type: Total Deformation
Unit: mm
Time: 1 s
28/09/2025 23:12:19
16.823 Max
14.954
13.085
11.216
9.3463
7.477
5.6078
3.7385
1.8693
0 Min
Max
Min